MS1Vtx_barrel_l2hcal

overview · all regions and selections · rank 71 / 169

rank 71 / 169 by endcap / ht40 / D_equal

region	selection	D_equal	D_pooled	125/15	125/55	200/50	400/100	600/150	1000/275	n_bkg
endcap	all	0.311	0.309	0.323	0.310	0.316	0.311	0.299	0.312	34,970
endcap	ht40	0.381	0.378	0.393	0.379	0.385	0.380	0.368	0.382	7,390
endcap	bkg split	0.089	below 27,580 / above 7,390							
barrel	all	0.259	0.269	0.293	0.239	0.246	0.283	0.277	0.267	1,089
barrel	ht40	0.404	0.410	0.434	0.376	0.389	0.421	0.417	0.407	220
barrel	bkg split	0.190	below 869 / above 220							
all	all	0.296	0.297	0.316	0.274	0.295	0.303	0.292	0.297	36,059
all	ht40	0.367	0.368	0.387	0.346	0.367	0.375	0.364	0.369	7,610
all	bkg split	0.091	below 28,449 / above 7,610							

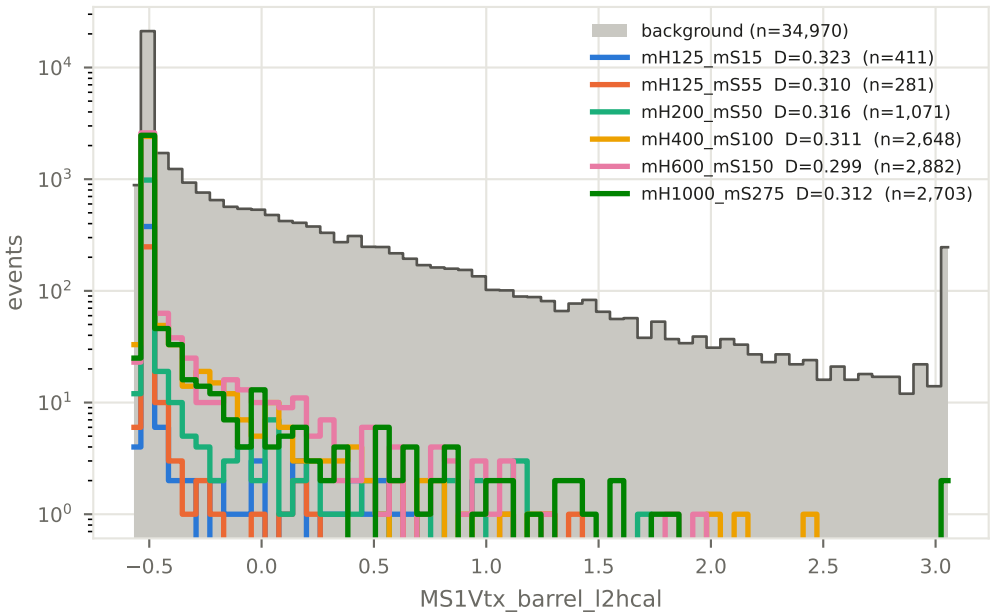
selections: all = every event; ht40 = htmissScalar_NOSYS >= 40 GeV, signal and background
pages: per region -> [all] 2x2, [ht40] 2x2, [background below vs above] 1x2

MS1Vtx_barrel_l2hcal

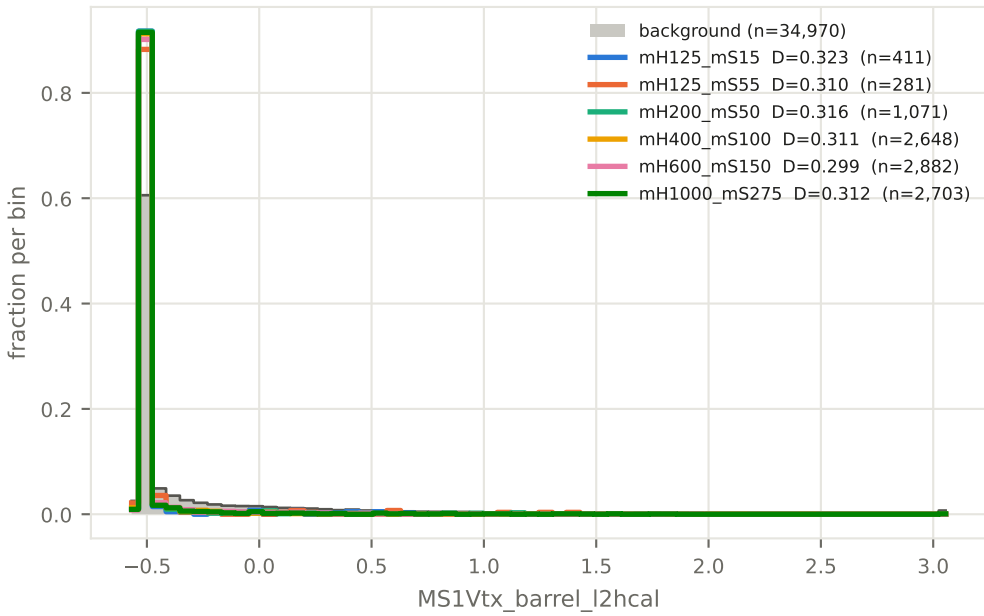
region: ENDCAP · selection: all (every event) · rank 71 / 169

D_equal 0.311 D_pooled 0.309 min 0.299 (mH600_mS150) max 0.323 (mH125_mS15) bkg n = 34,970 sig n = 9,996

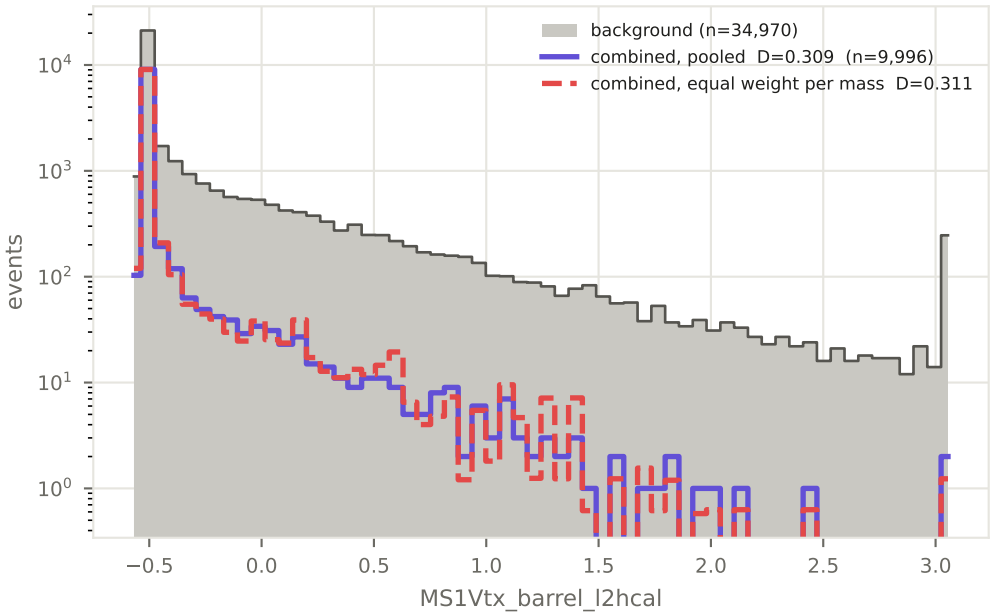
each mass point vs background · raw



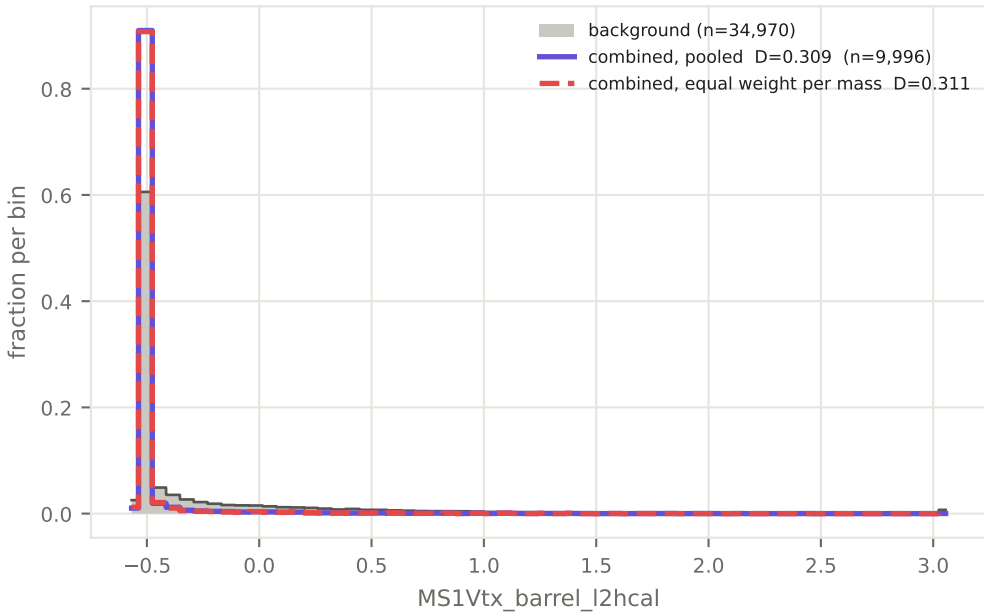
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



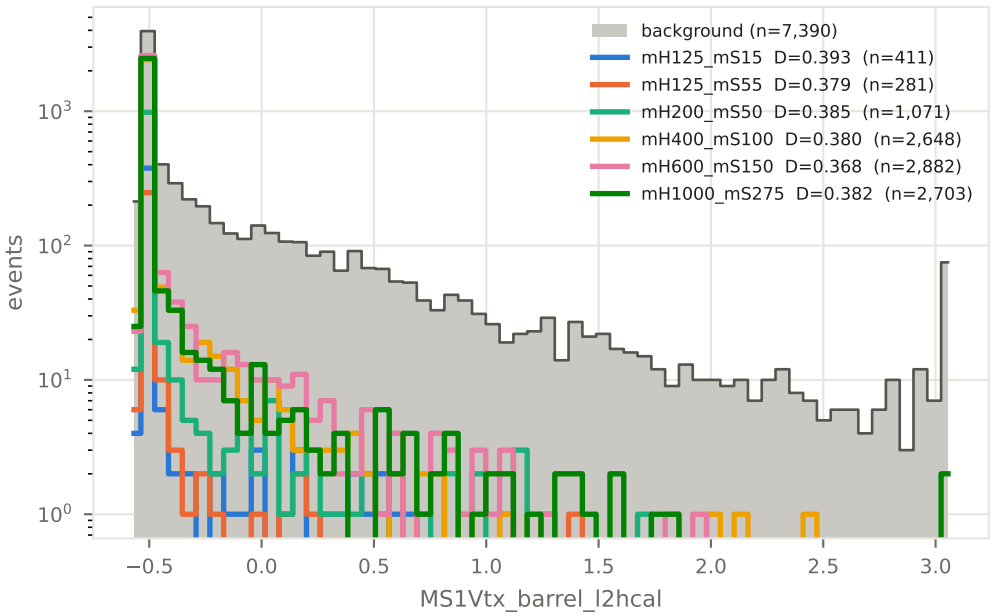
edge bins include under/overflow · NaN and -999 removed

MS1Vtx_barrel_l2hcal

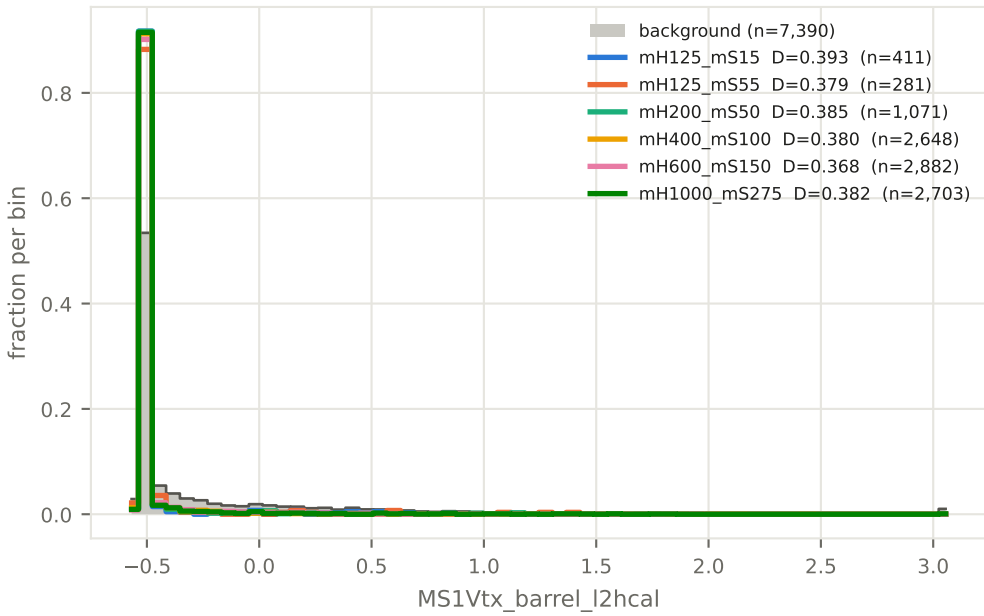
region: ENDCAP · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 71 / 169

D_equal 0.381 D_pooled 0.378 min 0.368 (mH600_mS150) max 0.393 (mH125_mS15) bkg n = 7,390 sig n = 9,996

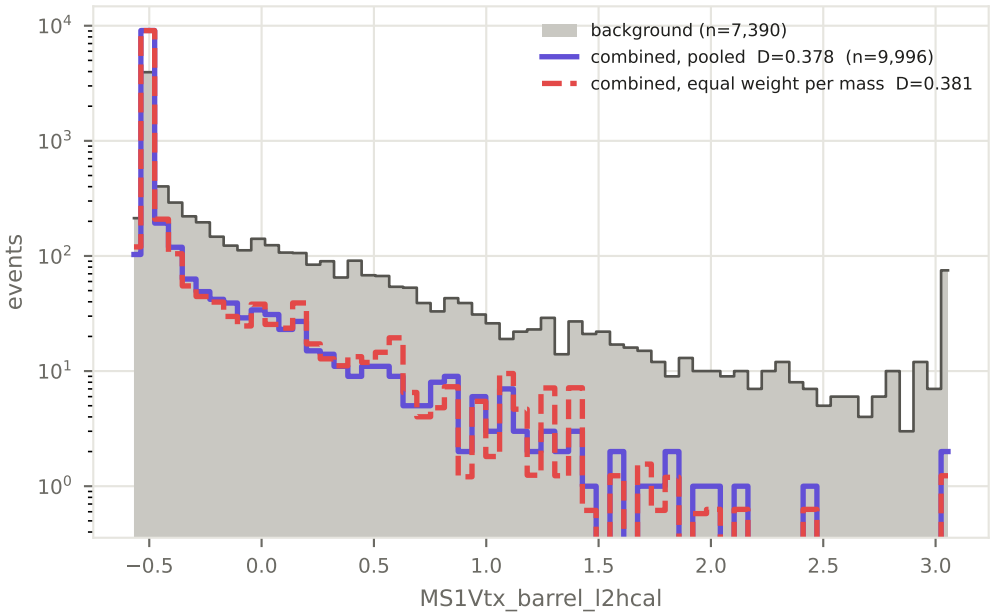
each mass point vs background · raw



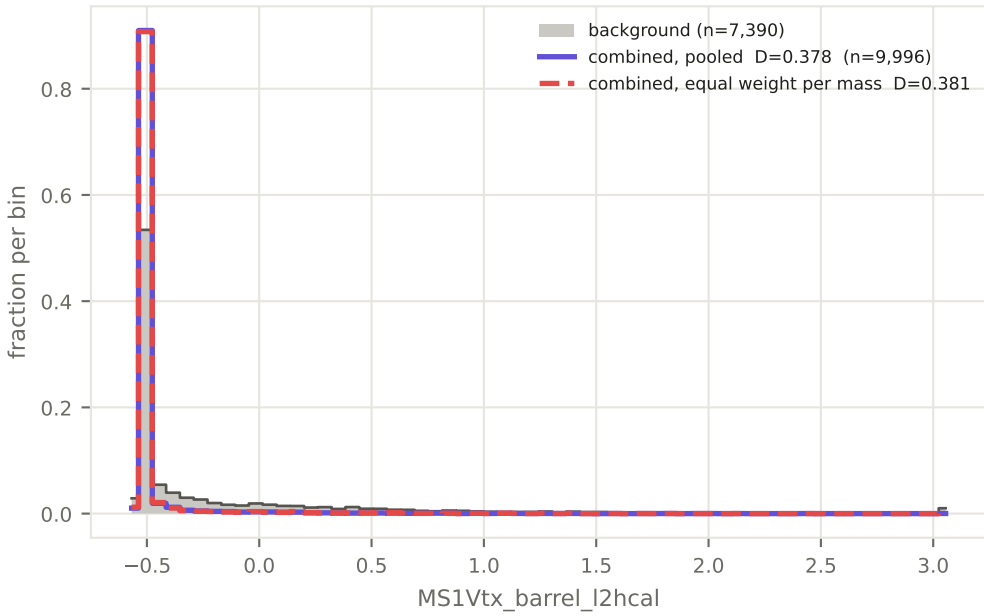
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized

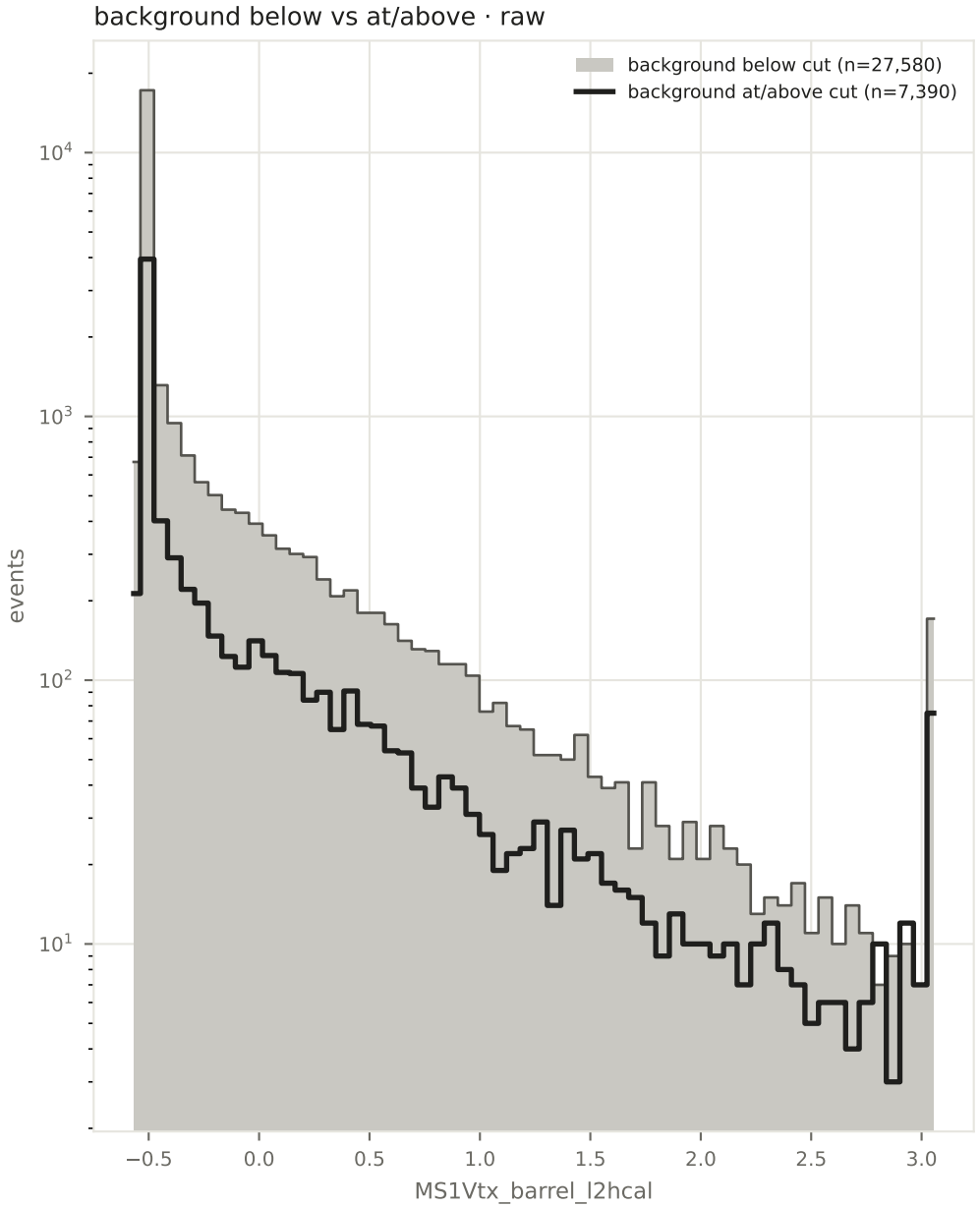


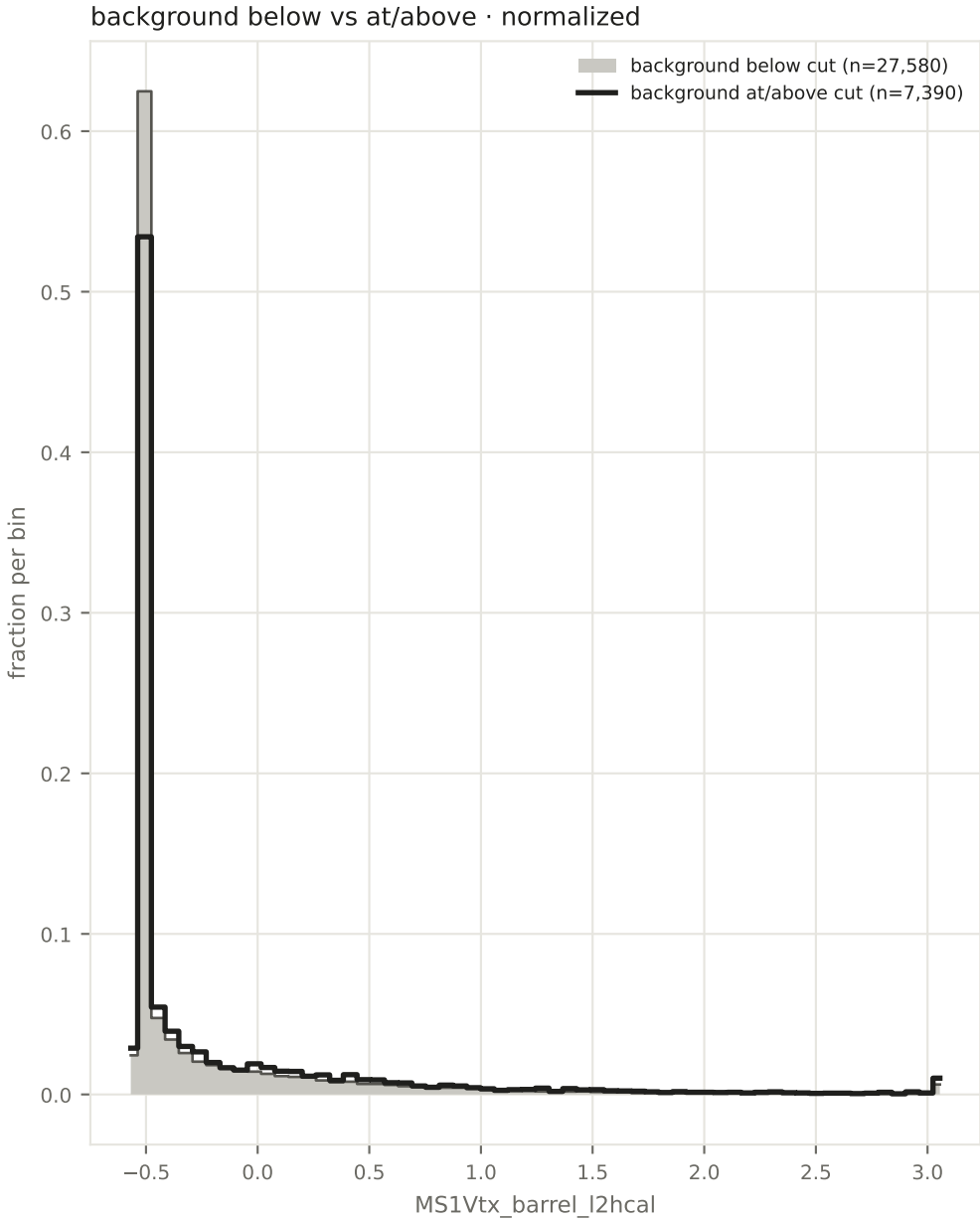
edge bins include under/overflow · NaN and -999 removed

MS1Vtx_barrel_l2hcal

region: ENDCAP · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 71 / 169

D(below vs above) 0.089 n below = 27,580 n above = 7,390 large D = background shape changes across the cut



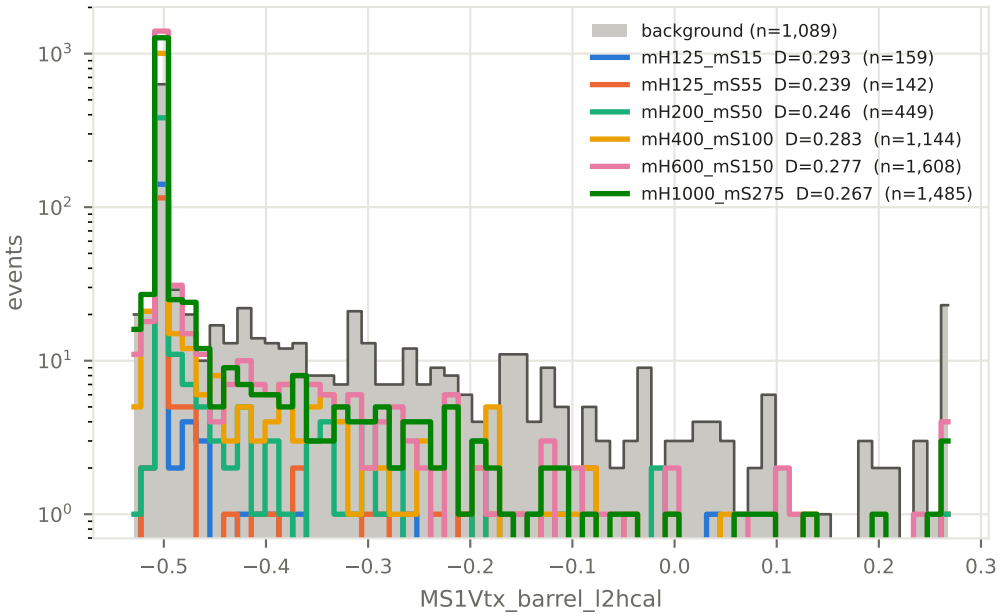


MS1Vtx_barrel_l2hcal

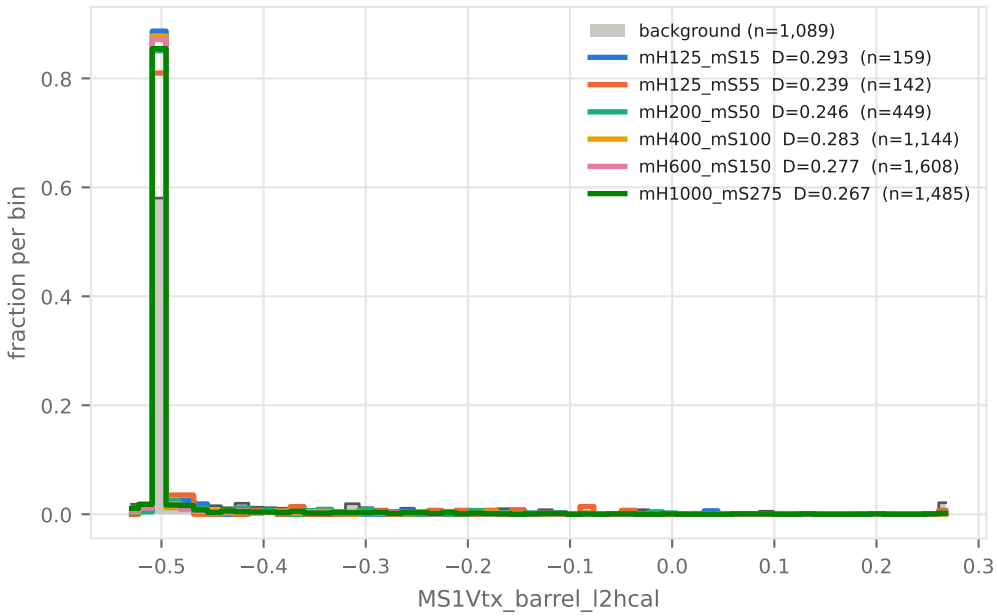
region: BARREL · selection: all (every event) · rank 71 / 169

D_equal 0.259 D_pooled 0.269 min 0.239 (mH125_mS55) max 0.293 (mH125_mS15) bkg n = 1,089 sig n = 4,987

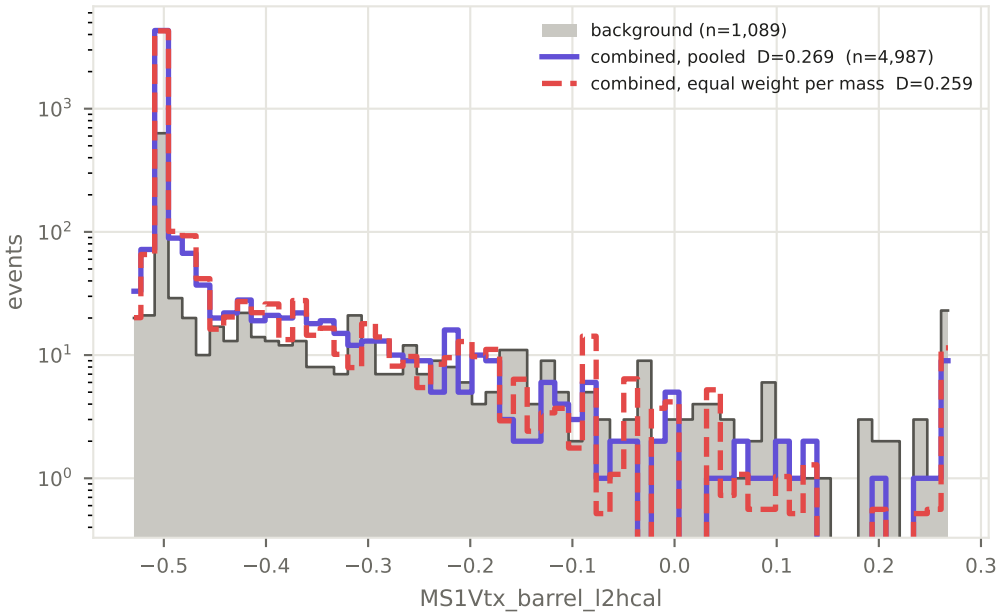
each mass point vs background · raw



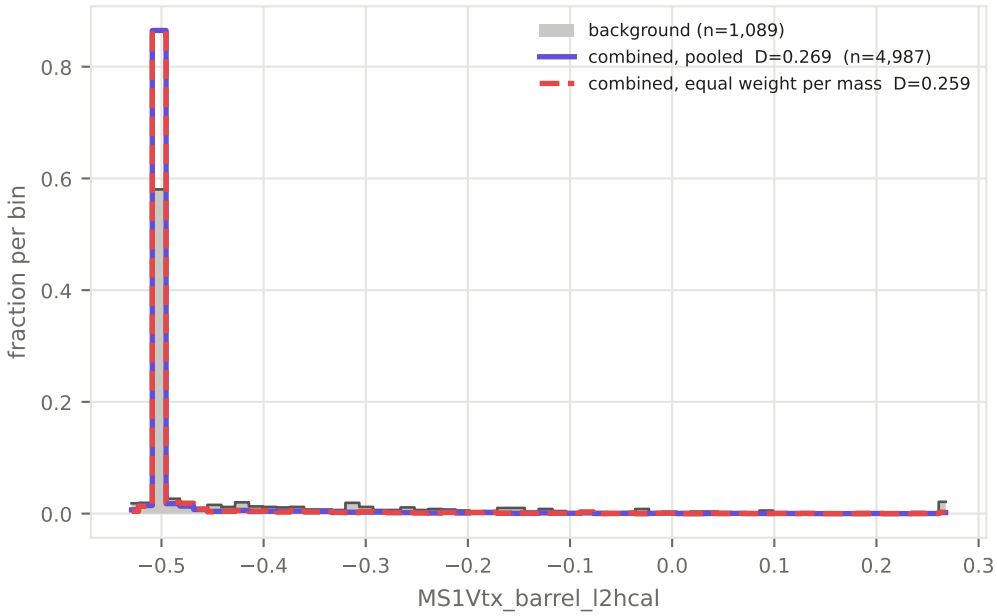
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



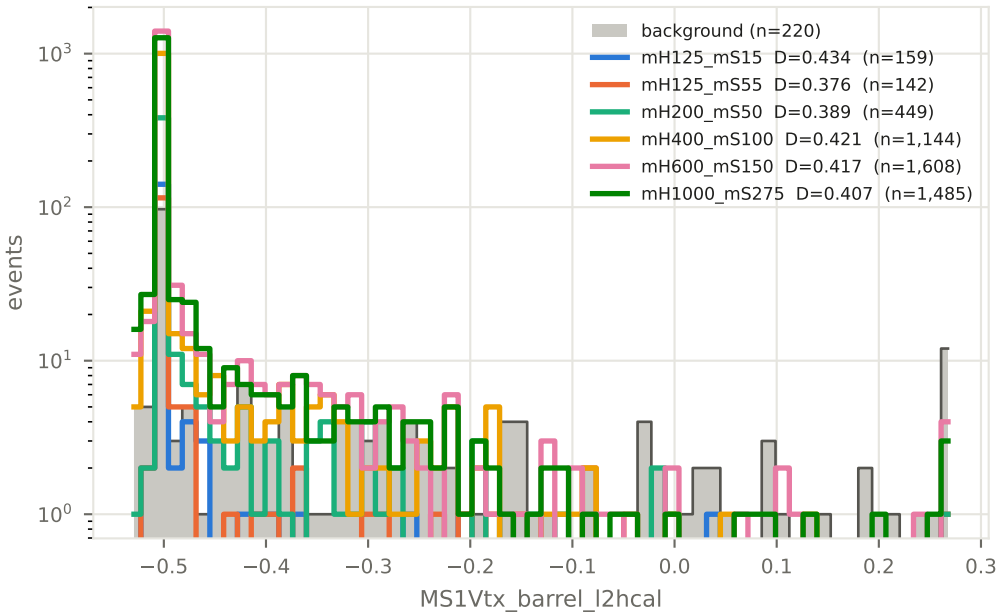
edge bins include under/overflow · NaN and -999 removed

MS1Vtx_barrel_l2hcal

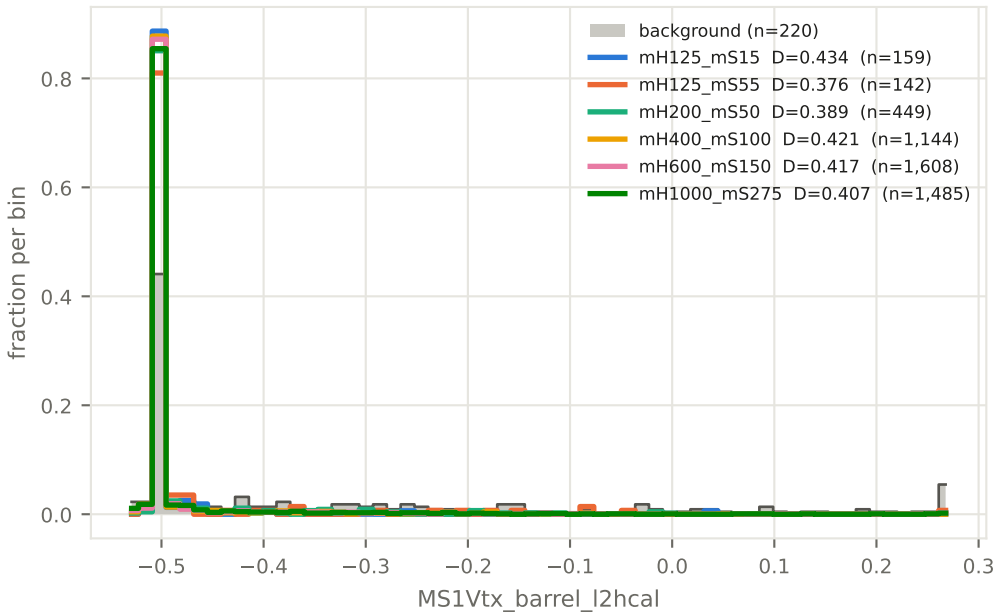
region: BARREL · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 71 / 169

D_equal 0.404 D_pooled 0.410 min 0.376 (mH125_mS55) max 0.434 (mH125_mS15) bkg n = 220 sig n = 4,987

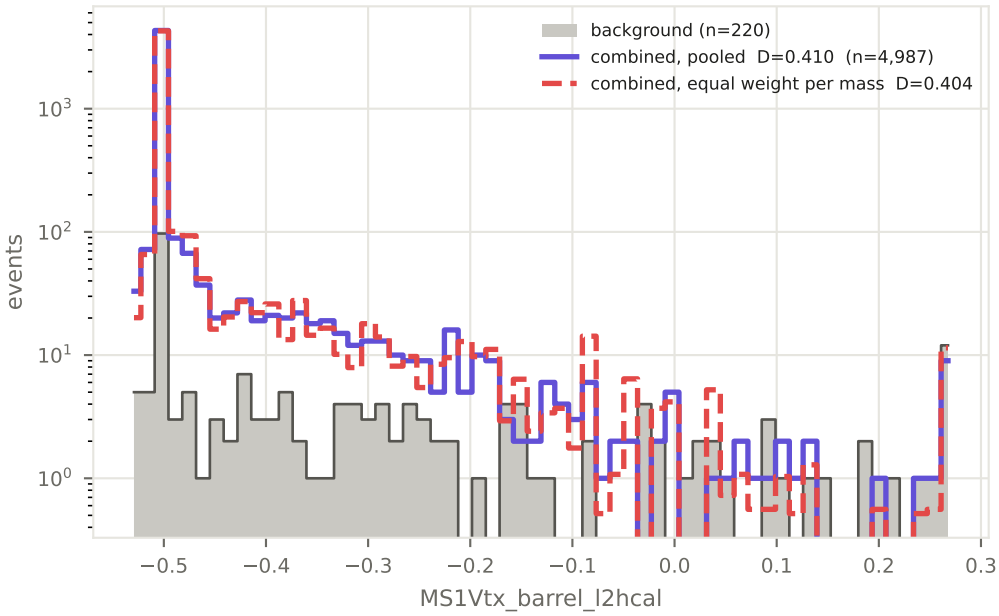
each mass point vs background · raw



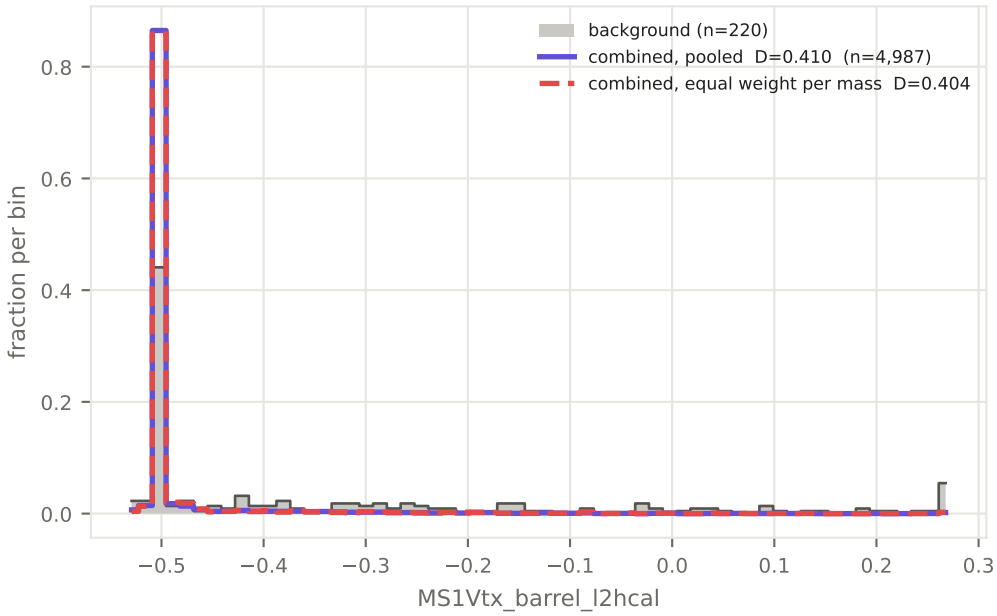
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized

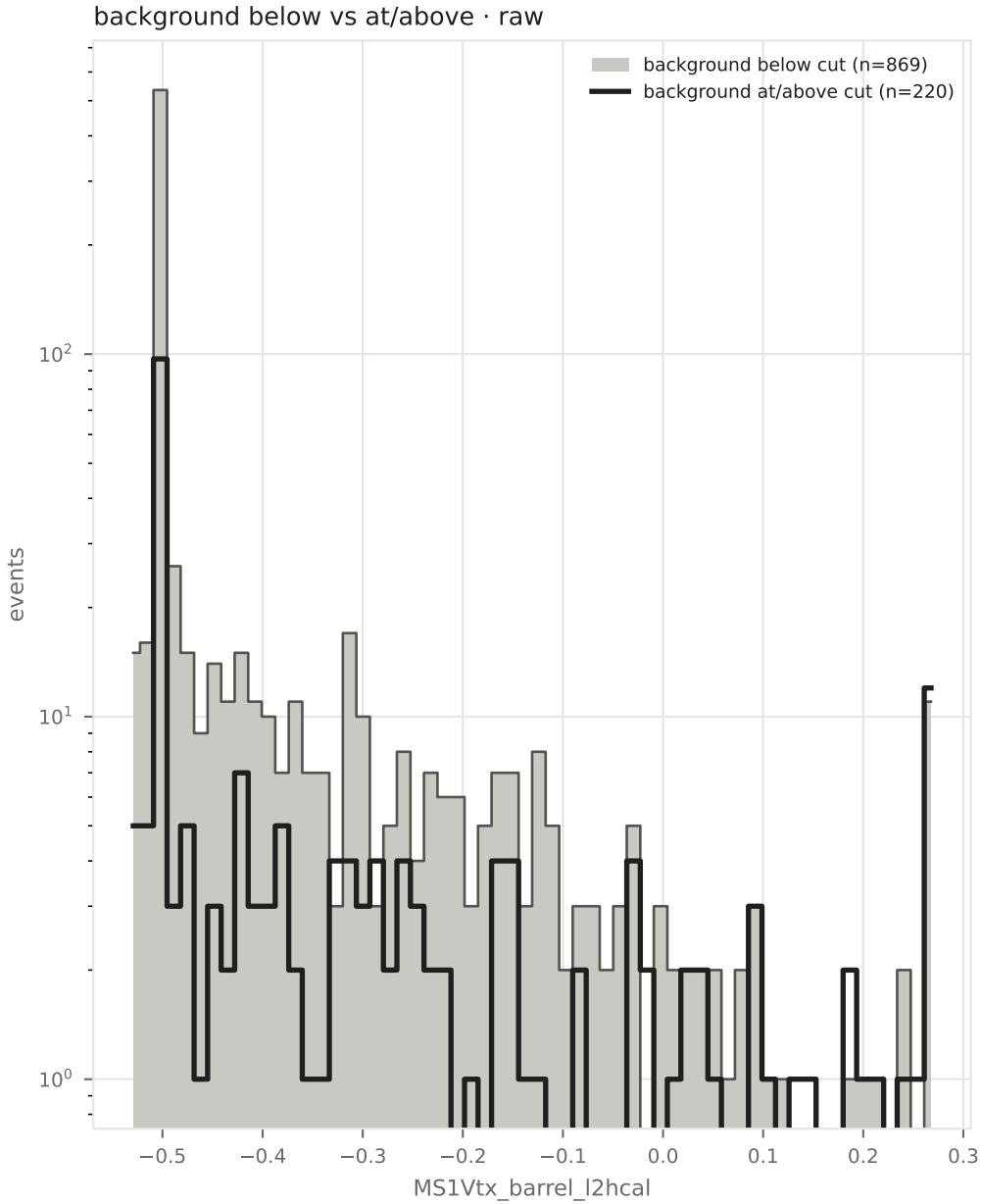


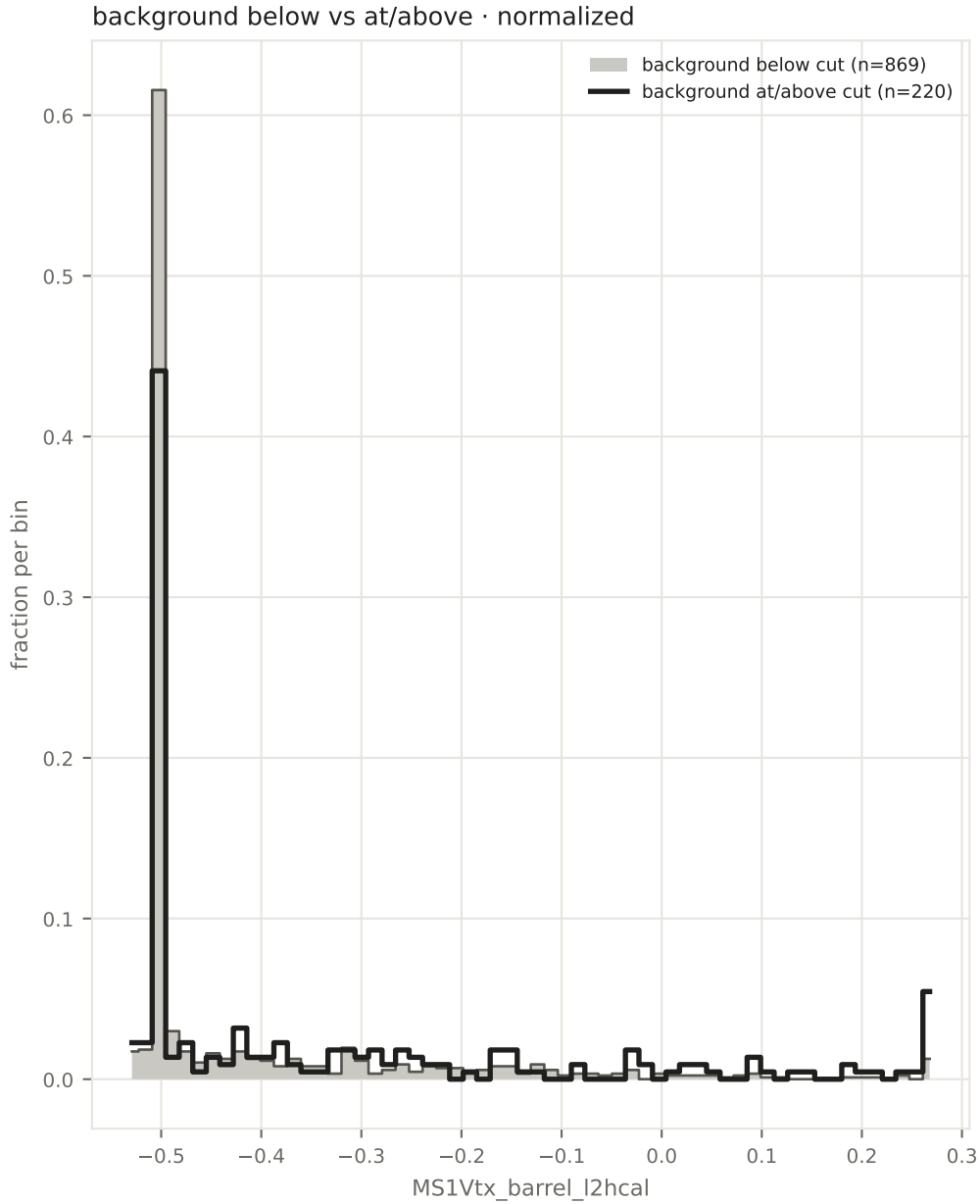
edge bins include under/overflow · NaN and -999 removed

MS1Vtx_barrel_l2hcal

region: BARREL · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 71 / 169

D(below vs above) 0.190 n below = 869 n above = 220 large D = background shape changes across the cut



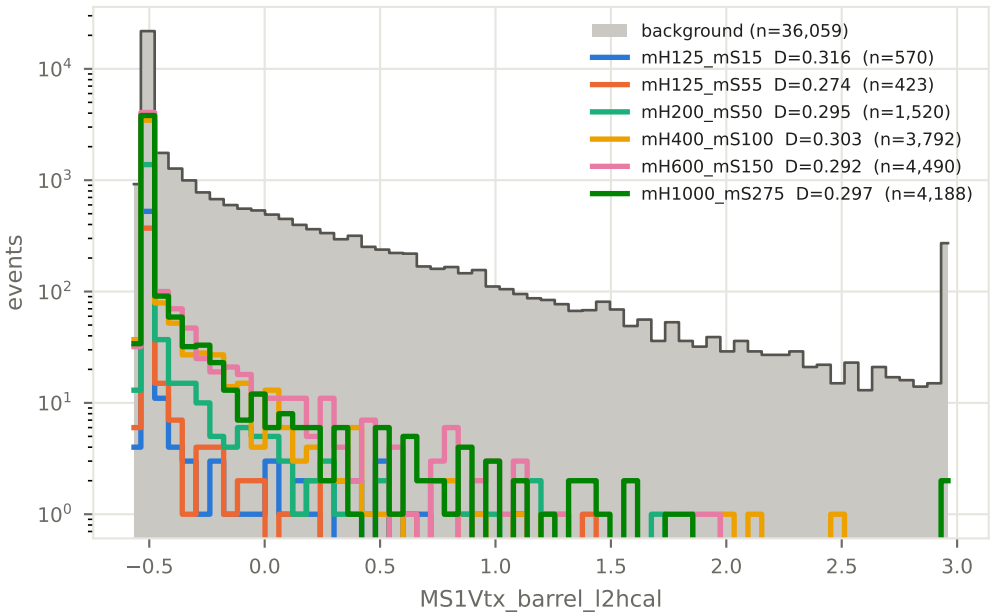


MS1Vtx_barrel_l2hcal

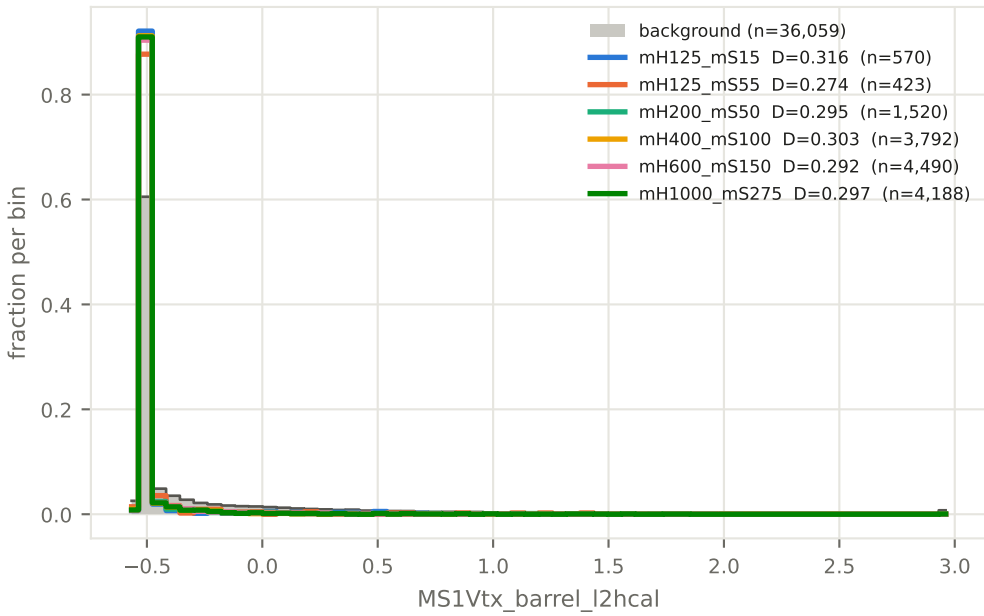
region: ALL · selection: all (every event) · rank 71 / 169

D_equal 0.296 D_pooled 0.297 min 0.274 (mH125_mS55) max 0.316 (mH125_mS15) bkg n = 36,059 sig n = 14,983

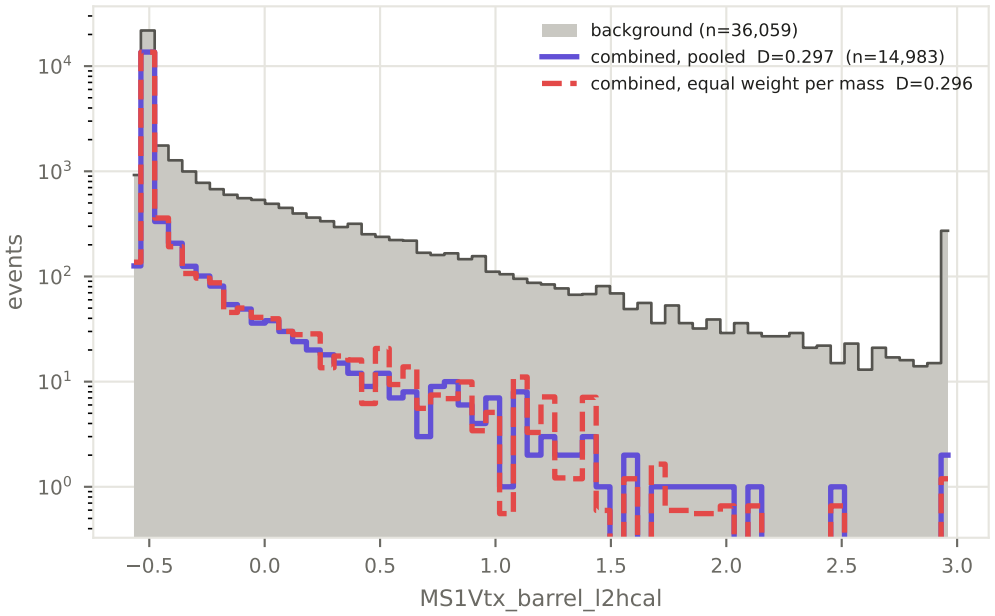
each mass point vs background · raw



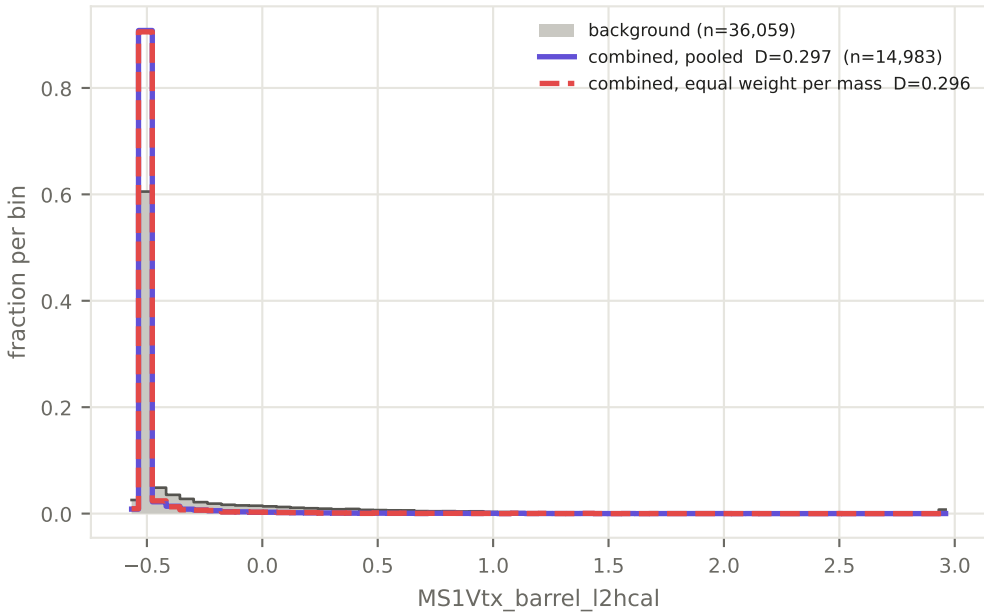
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



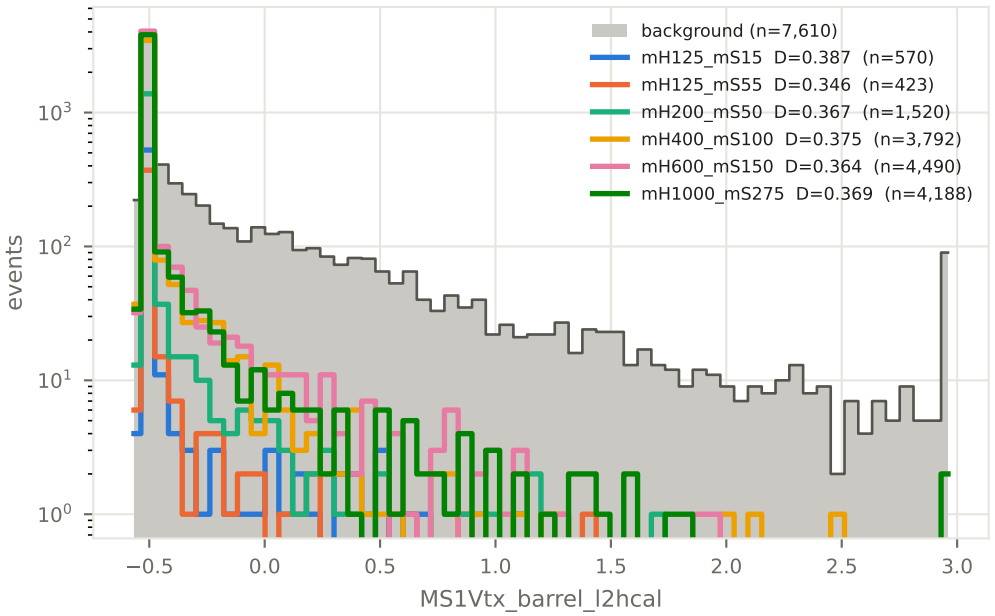
edge bins include under/overflow · NaN and -999 removed

MS1Vtx_barrel_l2hcal

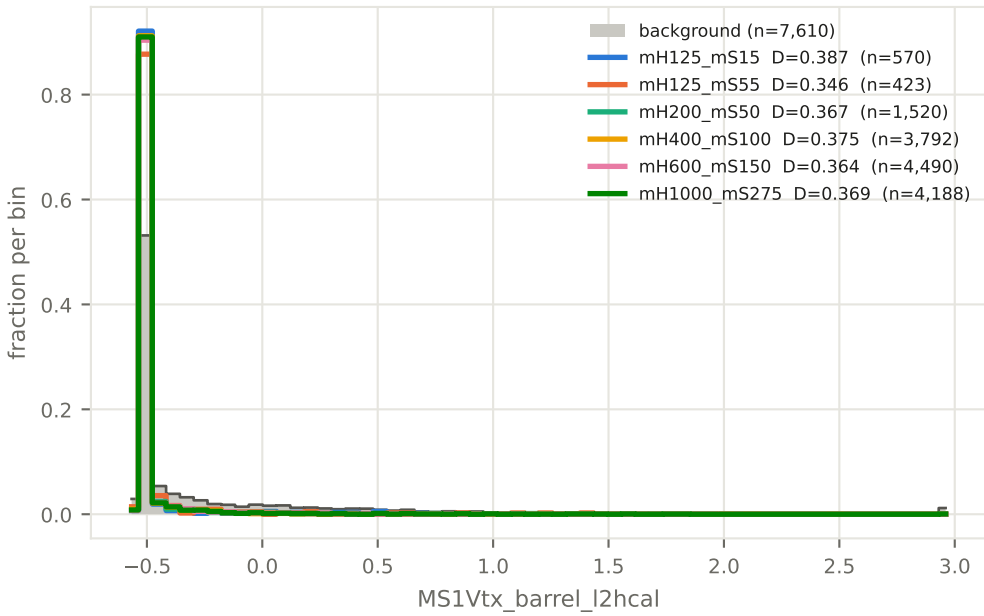
region: ALL · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 71 / 169

D_equal 0.367 D_pooled 0.368 min 0.346 (mH125_mS55) max 0.387 (mH125_mS15) bkg n = 7,610 sig n = 14,983

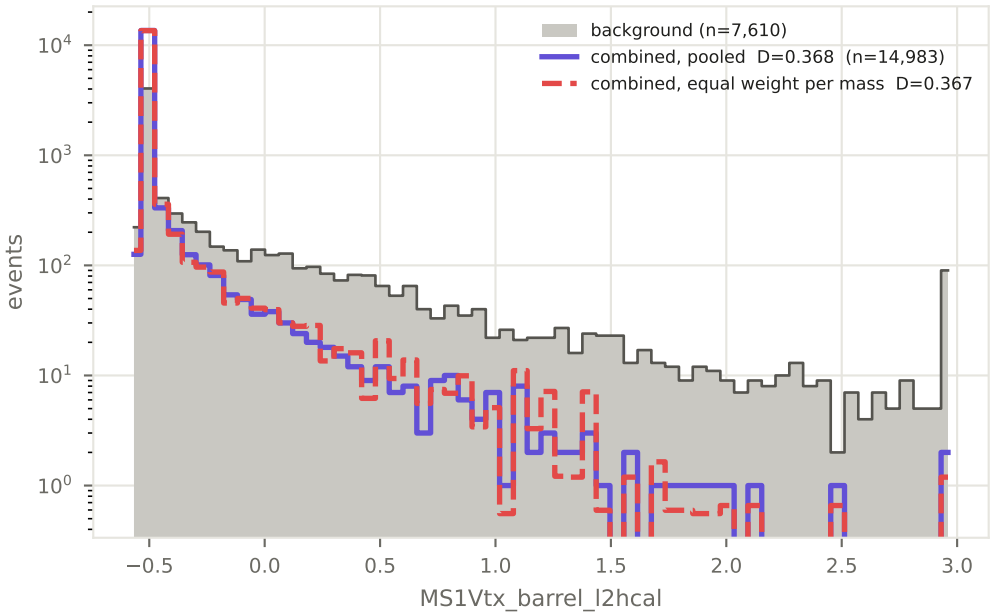
each mass point vs background · raw



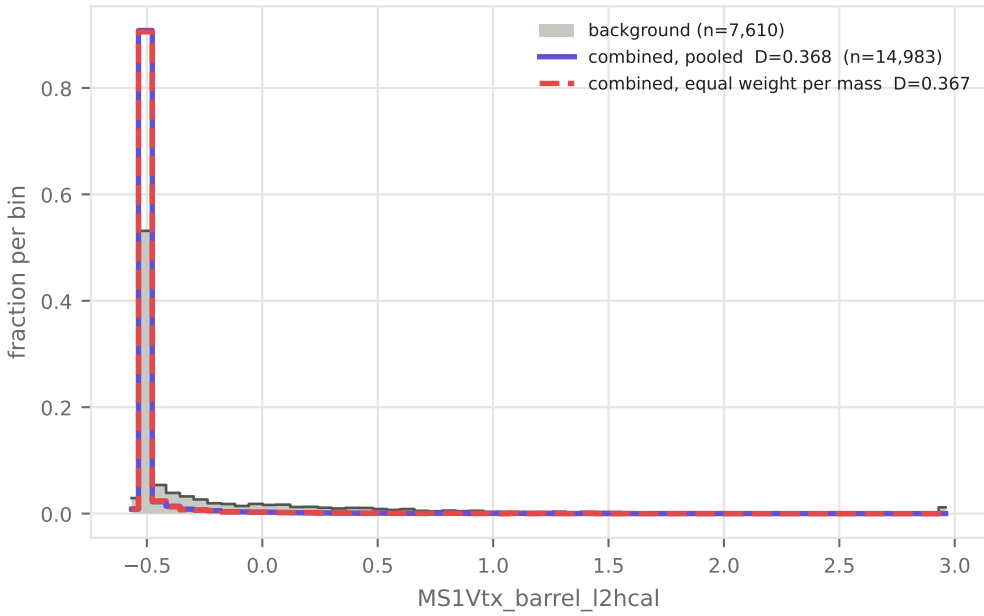
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



edge bins include under/overflow · NaN and -999 removed

MS1Vtx_barrel_l2hcal

region: ALL · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 71 / 169

D(below vs above) 0.091 n below = 28,449 n above = 7,610 large D = background shape changes across the cut

